Counting and Sampling Complexity of Total Matchings

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Abstract

The computational complexities of counting independent sets and matchings in graphs have been extensively studied in both exact and approximate regimes; however, the problem of counting total matchings; a structure uniting constraints of both independent set and matching remains largely unexplored. In this dissertation, we establish the hardness of exact counting for #TOTAL MATCHINGS in general graphs. We then investigate approximate counting in bounded-degree graphs using moment analysis and thermodynamic methods inspired by the study of the hardcore model. Based on analysis introduced by Sly, Weitz, Mossel and Wormald we derive the first moment rate function, the corresponding stationary point equations, and a local stability criterion for the symmetric solution. Numerical analysis indicates that the structure of the stationary point equations changes as the density of matching edges varies, suggesting a phase transition phenomenon. We also discuss the difficulties involved in converting this structural analysis into a rigorous approximation hardness reduction.

Research Ethics Approval

This project was planned in accordance with the Informatics Research Ethics policy. It did not involve any aspects that required approval from the Informatics Research Ethics Committee.

Declaration

I declare that this thesis was composed by myself, that the work contained herein is my own except where explicitly stated otherwise in the text, and that this work has not been submitted for any other degree or professional qualification except as specified.

(*Vinit Kate*)

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Chapter 1

Introduction

A total matching in a graph is defined as a disjoint union of an independent set of vertices and a matching of edges, constrained such that no vertex in the independent set is incident to any edge in the matching. In this project, we explore the counting complexity of the total matchings problem. The objective of this work is to establish the hardness result for exact counting of total matchings and to investigate the structural and computational behaviour of approximate counting in bounded-degree graphs. We will first investigate the complexity for exact counting in general graphs and then explore possible random structures to investigate the approximate counting complexity in bounded degree graphs.

1.1 Complexity Class #P

In complexity theory the class P consists of decision problems that can be solved in polynomial time by a deterministic Turing machine, and class NP consists of problems that can be verified in polynomial time given the problem instance and a certificate. The counting complexity classes corresponding to P/NP are FP/#P. The class #P was introduced by Leslie Valiant in 1979 [6].

For any decision problem $A \in NP$ we will consider the corresponding search problem for A that computes the solution for A, then #A denotes the version of the problem that computes the total number of valid solutions to A. Interestingly, this relationship does not trivially mirror the P versus NP landscape. For any decision problem $A \in NP$ the corresponding counting version, $\#A \in \#P$ but this is not true for class P, as for some of the problems in P, the corresponding counting version of those problems lies in $\#P$. For example, counting perfect matchings in a bipartite graph (#PerfMatch), counting

satisfying assignments for 2-variable-per-clause constraint formulas (#2SAT).

As most exact counting problems are in #P, there has been significant focus on approximate counting of the problems or by relaxing some criterion by which the problem becomes tractable. The class of algorithms that compute approximate solutions to a counting problem in polynomial time are known as the Fully Polynomial Randomised Approximation Scheme i.e. FPRAS. An FPRAS is an algorithm that returns a solution within a $(1+\varepsilon)$ multiplicative factor of the true count, running in time polynomial in the size of the input (n) and $\frac{1}{\varepsilon}$. To prove a given problem is hard to approximate, we will use approximation preserving reductions (AP-reductions), that preserves the existence of an FPRAS between problems.

A property we rely on in this project is the connection between approximate counting and approximate sampling. For a wide class of problems known as self-reducible problems, where the solution space of a large instance can be recursively partitioned by fixing components of the structure, such as including or excluding a specific vertex, approximating the total count of valid solutions is polynomially equivalent to sampling a solution almost uniformly at random from the configuration space. This equivalence, established by Jerrum, Valiant, and Vazirani [3], implies that the existence of a Fully Polynomial Randomised Approximation Scheme (FPRAS) for computing a partition function strictly guarantees the existence of a Fully Polynomial Almost Uniform Sampler (FPAUS). For related self reducible counting problems, the relationship between approximate counting and approximate sampling provides an important route to approximation hardness proofs. We use this framework as motivation for the possible hardness reductions in Chapter 3.

1.2 Previous Work: #INDSET, #MATCHINGS

The complexities of exactly and approximately counting independent sets have been studied in extensive detail. Because this problem corresponds to the hardcore model in statistical physics, these connections have motivated the extension of results to prove computational hardness. It is a well-established fact that exact counting of independent sets (#INDSET) is #P-hard, even when restricted to graphs with a maximum degree of $\Delta \geq 3$ [4].

Dyer, Frieze, and Jerrum proved that there is no FPRAS for independent sets for graphs of maximum degree $\Delta \geq 25$ unless NP = RP [1]. Sly further improved this result showing that for any graph of maximum degree $\Delta \geq 6$ and $\lambda = 1$ there is no

FPRAS for independent sets [5], in this paper Sly showed analysis using the hardcore model of lattice gas systems, and it was shown that when λ is slightly greater than the critical value, there is no FPRAS for the #INDSET problem for degree $\Delta \geq 3$. This breakthrough in approximation hardness was achieved by utilising random bipartite graphs as reduction gadgets.

Similarly the counting of matchings has been studied in detail, but the problem of counting total matchings remains to be explored. In this project we prove that the #TOTAL MATCHINGS in general graphs is #P-hard. We also replicate the analysis by Weitz, in which he introduced a novel analysis based on decay of correlation on regular trees with distance, proving the uniqueness of critical activity, which also happens to be the region of existence of the approximation scheme for independent sets [7]. We use similar analysis to explore the existence of FPRAS for counting of total matchings.

1.3 A Trivial Positive Result and Outline of The Project

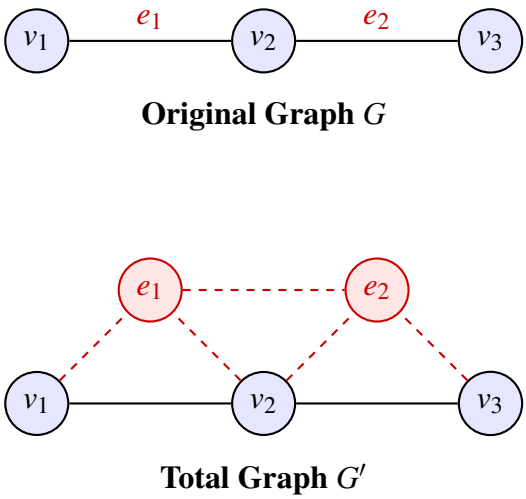


Figure 1.1: A linear graph G and its total graph G'. In G', the original edges e_1 and e_2 are converted into vertices (red). The dashed lines illustrate the new adjacencies formed by vertex-edge incidences and edge-edge adjacencies from G.

It can be observed that a total matching in a graph G directly corresponds to an independent set in the total graph of G. The total graph G' for a given graph G is constructed such that its vertex set consists of all vertices and all edges of G. In G', two vertices are adjacent if their corresponding elements in G are adjacent or incident (i.e., two vertices connected by an edge, two edges have the same incident vertex, or a vertex

and its incident edge). The maximum degree of any vertex in G' is twice the maximum degree of G.

This structural mapping implies that exact counting algorithms for total matchings exist trivially for graphs of maximum degree 2, since #INDSET is known to be hard for all graphs with degree $\Delta \geq 3$. [4]. It is also easy to see that in such graphs total matchings can be counted exactly by setting up recursive functions. Because counting total matchings encompasses counting independent sets as a subproblem, it can be guessed that the problem is at least as hard as counting independent sets.

- In chapter 2, we start by defining several relevant problems, introducing some new terms that will be useful in proving the hardness of #TOTAL MATCHINGS problem in general graphs. We prove that the exact counting of total matchings parameterised by the size of an independent set and the size of the matchings is #P-hard. This is achieved using a multicall reduction from #INDSET via polynomial interpolation. We then extend this to prove that exact counting of total matchings is also hard via polynomial interpolation. Then we prove two lemmas showing the coefficient matrix (formed by the system of equations in proof of theorem 2) is invertible.

- In chapter 3, we investigate the hardness of approximate counting for total matchings in bounded degree graphs, building upon the thermodynamic phase transition analysis demonstrated by Sly, Mossel, Weitz, and Wormald [4, 5].

- We conclude our work in chapter 4, also including possible extensions and improvements.

Chapter 2

Hardness of Counting Total Matchings in General Graphs

In this chapter, we establish that computing the number of total matchings in an arbitrary graph is a hard problem. Theorem 2 is our main result, which proves that exact counting of total matchings (#TOTAL MATCHINGS) is #P-hard. We also discuss the approximate counting problem and explain why extending the exact counting result to bounded degree graphs requires additional structural arguments. Adaptation of the MAX-CUT reduction as proved by Sly [5] for #INDSET, which builds on the hardcore model, needs to account for the matching constraint, since matching edges are globally dependent when they share vertices.

Here is an overview of the reductions that we follow to prove the hardness of the problem: We introduce an intermediate parameterised problem, #TOTAL MATCHINGS(i, j), which isolates the count of total matchings to a specific independent set size i and matching size j. The proof proceeds in two steps: we first reduce the known #P-hard problem i.e. #INDSET to this intermediate parameterised version of the problem, and then we prove the hardness of the #TOTAL MATCHINGS problem by interpolating it from the intermediate problem.

Before detailing the formal hardness proofs, we introduce the definitions and notation used throughout this chapter.

Definition 1. (Independent Set) Given a graph $G(V,E)$, a subset of vertices $V' \subseteq V$ forms an independent set if no two vertices in V' are adjacent. For any pair of vertices $u, v \in V'$, the edge $\{u,v\} \notin E$.

Definition 2. (Matching) Given a graph $G(V,E)$, a subset of edges $E' \subseteq E$ forms a matching if no two edges in E' share a common incident vertex.

Definition 3. (Total Matching) Given a graph G(V, E), any subset V' of vertices and E' of edges forms total matching if V' is an independent set, E' forms matchings, and there is no edge in E' which in incident to any vertex in V' i.e. no edge in E' covers any vertex from V'.

Definition 4. Let G(V, E) be a graph; then **#INDSET** is the counting problem of computing the total number of valid independent sets in G. #INDSET = $|\{V' : V' \subseteq V$ and V' forms independent set in G$\}|$

Definition 5. Let G(V, E) be a graph. **#MATCHINGS** is the counting problem of computing the total number of valid matchings in G. #MATCHINGS = $|\{E' : E' \subseteq E$ and E' forms matchings in G$\}|$

Definition 6. Let G(V, E) be a graph. **#TOTAL MATCHINGS(i, j)** is the parameterised counting problem of computing the total number of valid total matching (V', E') in G such that the independent set has size $|V'| = i$ and the matching has size $|E'| = j$.

2.1 Proof Method: Polynomial Interpolation

To establish computational hardness for exact counting problems, we employ the technique of polynomial interpolation. The core objective of this method is to extract a specific, computationally hard combinatorial quantity that is included within a global sum.

In the context of counting reductions, we construct a Turing (multicall) reduction by assuming access to a hypothetical polynomial-time oracle for our target problem (in this case, #TOTAL MATCHINGS). If we can utilise a known #P-hard problem such that its solution corresponds to a specific isolated term within the global partition function evaluated by this oracle, we can prove that the target problem is at least as hard as the initial problem.

A single query to the target oracle is usually insufficient to solve the problem, as the oracle returns an aggregated sum of many unknown structural combinations. To isolate the desired term, we must force the oracle to yield multiple distinct aggregated sums within the same combinatorial space. We achieve this by cleverly modifying the input graph through a family of polynomial-time structural transformations, typically parameterised by an integer k.

By querying the oracle for each modified graph (increasing k), we construct a system of linear equations. The total sums returned by the oracle form the constant vector,

while the unknown variables represent the structural counts of the original graph (such as #TOTAL MATCHINGS(i, j)). This forms a system of equations of the form $Mx = b$. If we can prove that this transformation matrix is non-singular (invertible), then the system of equations has a unique solution. Consequently, we can invert the matrix using standard linear algebra in polynomial time, isolate the exact count of the underlying #P-hard problem, and definitively conclude the hardness reduction.

2.2 Hardness Proofs

With this framework established, we now proceed to the formal hardness proofs. We structure our argument in two distinct stages. First, we demonstrate in Theorem 1 that the parameterised problem #TOTAL MATCHINGS(i, j) is #P-hard through a direct multi-call reduction from #INDSET. Building upon this, we deploy the polynomial interpolation technique described above in Theorem 2. By modifying the target graph and constructing an invertible system of equations, we utilise the hardness of the parameterised problem to prove that computing the #TOTAL MATCHINGS count is #P-hard.

2.2.1 Exact Counting

Theorem 1. #TOTAL MATCHINGS(i, j) is #P-hard.

Proof. To prove the hardness of the parameterised problem #TOTAL MATCHINGS(i, j), we we construct a polynomial-time Turing reduction from #INDSET, which is well known #P-hard problem.

Let $G(V,E)$ be an arbitrary graph with $|V| = n$ vertices. The #INDSET problem requires us to compute the total number of valid independent sets in G.

By Definition 6, #TOTAL MATCHINGS(i, j) evaluates the number of total matchings in G consisting of independent sets of size i and matchings of size j. If we restrict the parameter j to be strictly 0, the structural constraint requiring that no matching edge is incident to an independent set vertex becomes vacuously true. Consequently, #TOTAL MATCHINGS(i, 0) computes the total number of independent sets of size i within G.

Because the maximum possible size of an independent set in a graph with n vertices is n, the total number of independent sets in G can be computed by aggregating the

independent sets of every possible size. This relationship can be expressed as:

$$\#\text{INDSET} = \sum_{i=0}^{n} \#\text{TOTAL MATCHING(i, 0)}$$

Assuming access to a hypothetical polynomial-time oracle for #TOTAL MATCHINGS(i, j), we can compute the exact value of #INDSET by making exactly $n+1$ distinct queries to the oracle (one for each $i \in \{0, 1, \ldots, n\}$ and $j = 0$).

Since $n+1$ queries requires time polynomial in the size of the input graph, this constitutes a valid polynomial-time Turing reduction. Therefore, we conclude that #TOTAL MATCHINGS(i, j) is at least as hard as #INDSET, establishing its #P-hardness. □

Now we move onto #TOTAL MATCHINGS to prove that computing the number of total matchings is equally hard; we leverage the method of polynomial interpolation introduced in 2.1. By modifying the input graph, we demonstrate that the global #TOTAL MATCHINGS count can be expressed as a polynomial combination of the parameterised problem. The relation between these two problems allows us to construct an invertible system of linear equations, directly reducing the #P-hard parameterised problem to the #TOTAL MATCHINGS problem, as formalised in Theorem 2. It is not trivial that the system of equations is invertible. We prove that the formed system of equations is invertible, i.e., the coefficient matrix has full column rank in Lemma 2.

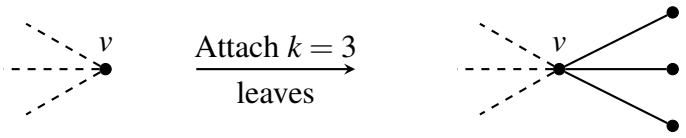
Theorem 2. #TOTAL MATCHING is #P-hard.

Proof. From the theorem 1, we have established that #TOTAL MATCHING(i, j) is #P hard.

Let $G(V,E)$ be an arbitrary graph with $|V| = n$ and $|E| = m$. We construct a family of modified graphs, denoted as G_k, by attaching exactly k new leaf vertices (pendant edges) to every original vertex $v \in V$ as demonstrated in figure 2.1. The maximum degree of any vertex in G_k is thus increased by k relative to its degree in G.

We can express the number of total matchings in G_k as a combinatorial sum over the parameterised counts #TOTAL MATCHINGS(i, j) of the original graph G. The valid cases for counting total matchings of the modified graph are (assuming size of the independent set is i and the size of the matching is j):

- If a vertex is covered by an independent set then none of its k neighbours can be included; the total number of such vertices are i. This gives total weight of 1^i.



Original Graph G Modified Graph G_3

Figure 2.1: For any vertex v, of G. The original graph G is transformed by attaching $k = 3$ leaf vertices to every vertex v.

- If a vertex is covered by matching, then any of its k neighbours can be included in an independent set. Thus each such vertex of the original graph contributes weight of 2^k the total number of such vertices are $2j$ giving a total weight of 2^{2jk}.
- If a vertex v is free, then possible ways for its k neighbours to be included in total matching are:
  1. Form a matching with v, then the rest of $k-1$ can be included in an independent set, contributing a weight of $k2^{k-1}$.
  2. Independent set: contributing a weight of 2^k.

  Each such vertex contributes a weight of $(k+2)\cdot 2^{k-1}$ the total number of such vertices are $n-i-2j$ giving a total weight of $(2^{k-1}(k+1))^{n-i-2j}$.

By evaluating the valid assignments for the independent sets and matchings extended into the k pendant leaves, the count in G_k evaluates to:

$$\begin{aligned}\#\text{TOTAL MATCHING} &= \sum_i \sum_j \#\text{TOTAL MATCHINGS(i, j)} \cdot (2^{k-1}(k+2))^{n-i-2j} \cdot 1^i \cdot 2^{k2j} \\ &= \sum_i \sum_j \#\text{TOTAL MATCHING(i, j)} \cdot \left(\frac{k+2}{2}\right)^{n-i-2j} \cdot 2^{(n-i)k}\end{aligned} \tag{2.1}$$

where the sum iterates over all valid pairs (i, j) satisfying the structural constraints:

$$0 \le i \le n \text{ and } i+2j \le n \tag{2.2}$$

Let N be the total number of valid pairs (i, j) that satisfy this condition, noting that $N \approx n^2/4$. We construct a system of linear equations by querying a hypothetical polynomial-time oracle for #TOTAL MATCHINGS on the modified graphs G_k across distinct integer values of $k = 0, 1, 2, \ldots, R$. This forms a matrix equation of the form:

$$\vec{b} = M_{R\times N}\vec{x}$$

where $\vec{b}$ is the column vector of oracle responses for each G_k, $\vec{x}$ is the column vector of the unknown #TOTAL MATCHINGS(i, j) variables, and C is the coefficient matrix with entries defined by:

$$M_{k,(i,j)} = \left(\frac{k+2}{2}\right)^{n-i-2j} \cdot 2^{(n-i)k}$$

Now we claim that the coefficient matrix is invertible, and then we can evaluate the #TOTAL MATCHING(i, j) for any i, j. By evaluating the system for a sufficiently large number of queries $R = (N+1)^2$, we extract an extended system of equations. If we can demonstrate that the columns of M are linearly independent, we can select N linearly independent rows to form a non-singular $N \times N$ sub-matrix.

Inverting this square matrix in polynomial time would allow us to isolate the exact count for #TOTAL MATCHINGS(i, j). This implication dictates that evaluating the #TOTAL MATCHINGS is at least as hard as the parameterised version, this completes the #P-hardness proof for exact counting.

To finalise the reduction, we now present two lemmas to prove our claim that the formed coefficient matrix M is invertible.

□

Lemma 1. Let $F(k) = \sum_{c=0}^{m} P_c(k)e^{\gamma_c k}$ be a non-zero quasi polynomial, where each polynomial $P_c(k)$ is of degree d_c and the γ_c are distinct real numbers. The number of real roots of $F(k)$, counted with multiplicity, cannot exceed:

$$R \leq \sum_{c=0}^{m} (d_c + 1) - 1 \tag{2.3}$$

Proof. The proof of this lemma is by induction on m.

Base Case ($m = 0$): The function consists of a single term $F(k) = P_0(k)e^{\gamma_0 k}$. Since the exponential function $e^{\gamma_0 k}$ is strictly positive and never evaluates to zero for any real k, the roots of $F(k)$ are entirely determined by the roots of the polynomial $P_0(k)$. By the fundamental theorem of algebra, a non-zero polynomial of degree d can have at most d real roots. Evaluating the upper bound formula for $m = 0$ yields $(d_0 + 1) - 1 = d_0$, which exactly matches this limit. Thus, the base case holds.

Inductive Step: Assume that the statement of the lemma is true for some integer $m \geq 0$. We must prove that it also holds for $m+1$; that is, any quasipolynomial of the form $F(k) = \sum_{c=0}^{m+1} P_c(k)e^{\gamma_c k}$ has at most $\sum_{c=0}^{m+1}(d_c + 1) - 1$ real roots.

To isolate the final polynomial term $P_{m+1}(k)$, we multiply the entire function by

$e^{-\gamma_{m+1}k}$ to define a new function $G(k)$:

$$G(k) = F(k)e^{-\gamma_{m+1}k} = P_{m+1}(k) + \sum_{c=0}^{m} P_c(k)e^{(\gamma_c-\gamma_{m+1})k}$$

Because $e^{-\gamma_{m+1}k} > 0$ for all $k \in \mathbb{R}$, this transformation scales the function but leaves the total number and location of its real roots strictly invariant. Let R_G denote the number of real roots of $G(k)$.

We now differentiate $G(k)$ exactly $d_{m+1}+1$ times with respect to k. Because the $(d_{m+1}+1)$-th derivative of a polynomial of degree d_{m+1} is identically zero, the isolated term $P_{m+1}(k)$ vanishes entirely. The resulting function, $Q(k) = G^{(d_{m+1}+1)}(k)$, takes the form:

$$Q(k) = \sum_{c=0}^{m} H_c(k)e^{(\gamma_c-\gamma_{m+1})k}$$

where each $H_c(k)$ is a newly formed polynomial. Because differentiating $P(k)e^{\alpha k}$ yields $(P'(k)+\alpha P(k))e^{\alpha k}$ and the exponential constants $(\gamma_c-\gamma_{m+1})$ are non-zero (since all γ_c are distinct), the degree of each polynomial $H_c(k)$ remains exactly equal to d_c.

Notice that $Q(k)$ is now a quasipolynomial with exactly $m+1$ terms (indexed from 0 to m). By our inductive hypothesis, the maximum number of real roots R_Q that $Q(k)$ can possess is bounded by:

$$R_Q \leq \sum_{c=0}^{m}(d_c+1) - 1$$

We now apply Rolle's Theorem, which states that "if a continuous, differentiable function has R real roots, its first derivative must possess at least $R-1$ real roots." Applying this theorem iteratively, differentiating a function D times reduces its minimum possible number of roots by at most D. Therefore, taking $d_{m+1}+1$ derivatives implies:

$$R_Q \geq R_G - (d_{m+1}+1)$$

Rearranging this inequality, we get:

$$R_G \leq R_Q + d_{m+1} + 1$$

Substituting the inductive bound for R_Q:

$$R_G \leq \left(\sum_{c=0}^{m}(d_c+1) - 1\right) + (d_{m+1}+1) = \sum_{c=0}^{m+1}(d_c+1) - 1$$

Because $F(k)$ and $G(k)$ share the exact same number of roots (R_G), we conclude that the maximum number of roots for $F(k)$ is strictly bounded by the summation over all $m+1$ terms.

□

Lemma 2. Let $G(V,E)$ be a graph with $|V| = n$. Let N denote the total number of valid structural pairs (i, j) satisfying $0 \leq i \leq n$ and $i + 2j \leq n$. The coefficient matrix M, defined by its entries $M_{k,(i,j)} = \left(\frac{k+2}{2}\right)^{n-i-2j} \cdot 2^{(n-i)k}$ for query indices $k \in \{0, 1, \ldots, (n+1)^2\}$, possesses full column rank N. Thus, there exists a strictly non-singular $N \times N$ submatrix that is invertible.

Proof. To prove that the coefficient matrix M is invertible, we will show that its columns are linearly independent.

Assume, for the sake of contradiction, that the columns are linearly dependent. Then there exists a set of scalar coefficients $A_{i,j}$, not all zero, such that a linear combination of the columns evaluates to zero for every row k:

$$f(k) = \sum_{i,j} A_{i,j} \cdot 2^{(n-i)k} \cdot \left(\frac{k+2}{2}\right)^{n-i-2j} = 0 \quad \forall k \in \{0, 1, 2, \ldots, (n+1)^2\}$$

To simplify the summation, we introduce the substitution $c = n - i$. Because $0 \leq i \leq n$, the new index c iterates from 0 to n. Grouping the terms by c gives a function $F(k)$:

$$F(k) = \sum_{c=0}^{n} 2^{kc} \left[\sum_{j} A_{n-c,j} \cdot \left(\frac{k+2}{2}\right)^{c-2j}\right] = 0$$

Let the following expression be denoted as $P_c(k)$:

$$P_c(k) = \sum_{j} A_{n-c,j} \cdot \left(\frac{k+2}{2}\right)^{c-2j}$$

$P_c(k)$ is a polynomial in k. The structural constraint $i + 2j \leq n$ algebraically implies $n - c + 2j \leq n$, which simplifies to $c - 2j \geq 0$. Therefore, the maximum exponent of k in $P_c(k)$ is bounded by c. Since $c \leq n$, the maximum degree of any polynomial $P_c(k)$ is at most n.

Substituting $P_c(k)$ back into the original sum gives a quasipolynomial:

$$F(k) = \sum_{c=0}^{n} P_c(k) \cdot 2^{kc} = 0$$

By Lemma 1, the maximum number of real roots R that this quasipolynomial can possess is strictly bounded above by:

$$R \leq \sum_{c=0}^{n} (\deg(P_c) + 1) - 1$$

Because $\deg(P_c) \leq n$, we evaluate the bound as:

$$R \leq \sum_{c=0}^{n} (n+1) - 1 = (n+1)(n+1) - 1 = (n+1)^2 - 1$$

However, our matrix construction forces $F(k) = 0$ at exactly $(n+1)^2 + 1$ distinct evaluation points (since k ranges inclusively from 0 to $(n+1)^2$). Because we have forced more roots than the theoretical maximum limit, the function $F(k)$ must be identically zero for all real numbers $k \in \mathbb{R}$.

Since $F(k) = 0$ everywhere, we can analyse its asymptotic limits. Assume for contradiction that not all polynomials $P_c(k)$ are the zero polynomial. Let c_{max} be the highest index where $P_{c_{max}}(k)$ is not identically zero. Factoring out the dominant exponential term $2^{kc_{max}}$ gives:

$$P_{c_{max}}(k) + \sum_{c<c_{max}} P_c(k) \cdot 2^{k(c-c_{max})} = 0$$

Taking the limit as $k \to \infty$, the exponent $(c - c_{max})$ is strictly negative for all terms in the summation. The exponential decay of $2^{k(c-c_{max})}$ goes to zero faster than any polynomial growth of $P_c(k)$, driving the entire summation term to zero. This leaves:

$$\lim_{k\to\infty} P_{c_{max}}(k) = 0$$

Since $P_{c_{max}}(k)$ is a non-zero polynomial in k, its magnitude must diverge to $\pm\infty$ as $k \to \infty$, creating a contradiction. Therefore, the assumption fails and $P_c(k)$ must be identically zero for all $c \in \{0, \ldots, n\}$

Finally, we analyse the internal terms of $P_c(k) = 0$:

$$\sum_j A_{n-c,j} \cdot \left(\frac{k+2}{2}\right)^{c-2j} = 0$$

As the index j increments, the degree of the polynomial terms drops sequentially by exactly 2 (yielding terms of degree $c, c-2, c-4, \ldots$). Because every term in this expansion possesses a unique degree, the terms are linearly independent. The only way for such a polynomial to evaluate identically to zero is if every scalar coefficient is strictly zero. Therefore, $A_{i,j} = 0$ for all valid pairs (i, j). Because all coefficients $A_{i,j}$ are forced to be zero, the columns of the matrix M are linearly independent. The matrix possesses a full column rank of N. By selecting any N linearly independent rows from this system, we extract a non-singular $N \times N$ square matrix that can be inverted to isolate the target #TOTAL MATCHINGS(i, j) counts, completing the hardness reduction.

□

2.3 Overview of Approximate Counting

The exact-counting result established above does not immediately extend to approximate counting. For related models such as independent sets, approximation hardness have been established through thermodynamic phase transitions and reductions based on random bipartite graphs [4, 5, 7]. These methods rely on a careful analysis of distinguishable phases and on a reduction that uses the phases of different gadgets.

For total matchings, the presence of the matching constraint introduces an additional dependence between edges. Consequently, the interaction between two phase gadgets cannot be represented by independent single edge weights when several connecting edges share vertices. A rigorous approximation hardness reduction therefore requires a separate analysis of the partition function of the interaction gadget.

In this chapter we have proved the computational intractability of the counting total matchings problem, via a sequence of multicall reductions, polynomial interpolations, and matrix inversion arguments, we first proved that the exact counting of total matchings in general graphs is #P-hard. In chapter 3, we investigate the approximation counting and issues related to the reduction methods. We derive the stationary point equations governing the first moment rate function and study the local stability of the resulting symmetric solution. This provides a basis for identifying phase transition regimes and for development of a rigorous bounded-degree hardness reduction.

Chapter 3

Phase Transitions and Approximate Counting in Bounded-Degree Graphs

We have established in Chapter 2 that exact counting of total matchings is #P-hard. In this chapter, we investigate the structure of the total matchings, using techniques motivated by the analysis of independent sets building on the hardcore model. We first derive the expected number of total matchings with densities of independent set vertices and matching edges. We then derive the first moment rate function and the corresponding stationary point equations. Finally, we study the local stability of the symmetric stationary point and numerically investigate the regimes in which the stationary point equations exhibit bifurcation.

The purpose of this analysis is to identify phase transition regimes. Establishing that such a transition implies NP-hardness of approximate counting following the work of Sly, which requires gadget construction and an approximation-preserving reduction, which are left as an open direction.

3.1 Previous Work in Approximate Counting (#INDSET)

The Hardcore model is a statistical physics model parametrised by fugacity, $\lambda > 0$ which represents the likelihood of a vertex being occupied i.e. included in an independent set. The partition function Z_G, computes the weighted sum of independent sets.

$$Z_G = \sum_{I \subseteq V} \lambda^{|I|}$$

Thus when λ increses the approximation of the partition function becomes a hard problem as more weight is concentrated on the large independent sets. For bounded-

degree graphs, the computational complexity of approximating the partition function is not uniform; it is entirely governed by a thermodynamic phase transition. This transition occurs at a critical threshold (on the d-regular tree), defined as $\lambda_c = \frac{(\Delta-1)^{\Delta-1}}{(\Delta-2)^{\Delta}}$ [5].

Weitz proved that for any graph of maximum degree Δ, if the fugacity λ is strictly less than $\lambda_c(\Delta)$, there exists a deterministic Fully Polynomial Time Approximation Scheme (FPTAS) to compute the partition function [7]. This analysis is based on the decay of correlation this implies that fixing the state of a set of vertices (the boundary) has an exponentially decaying effect on vertices located far away from that boundary.

Mossel, Weitz, and Wormald showed that for d-regular bipartite graphs with $d \geq 3$ and λ just above λ_c, the Gibbs measure exhibits non-uniqueness [4]. In this region, strong spatial mixing fails. If one fixes the distant leaves of the tree into a specific configuration, that bias propagates to the root without decaying. This implies that local Markov chains take exponential time to get close to the stationary distribution in this region. This slow mixing demonstrates the failure of MCMC methods.

Sly proved that for $\lambda > \lambda_c(\Delta)$ (specifically for $\Delta \geq 6$, later generalised to all $\Delta \geq 3$ by Galanis, Štefankovič, and Vigoda [2]), no Fully Polynomial Randomised Approximation Scheme (FPRAS) can exist unless NP = RP [5]. Sly proved this by showing a reduction to MAX-CUT and random bipartite graphs as gadgets. We will use similar random structures and analysis of first-moment/second-moment for #TOTAL MATCHINGS.

For the hardcore model, the connection between spatial mixing, phase non-uniqueness, and approximation complexity motivates the use of similar methods for total matchings. In this dissertation, we use these methods to investigate the stationary point structure and local stability of the total matching model, following the approaches of Sly, Mossel, Weitz, and Wormald. Based on the analysis by Mossel, Weitz, and Wormald [4] they prove that a set of balanced independent sets in a random bipartite graph forms an exponentially small bottleneck, resulting in exponential mixing time of local Markov chains. We introduce one more parameter γ that represents the proportion of matchings, along with α, β representing proportions of vertices selected. The random bipartite graph gadget is sampled from a set of bipartite graphs obtained by taking d random perfect matchings between two sets, V_1, V_2 both of size n [4].

3.2 Moment Analysis

3.2.1 Computing Expected Value of $\mathcal{T}_G^{\alpha,\beta,\gamma}$

Let $G = (V_1 \cup V_2, E)$ be a random bipartite graph where the two partitions have equal size $|V_1| = |V_2| = n$. We define $\mathcal{T}_G^{\alpha,\beta,\gamma}$ as the set of all valid total matchings (I, M) in G that satisfy density parameters:

$$\mathcal{T}_G^{\alpha,\beta,\gamma} = \{(I,M) | I \cap V_1 = \alpha n, I \cap V_2 = \beta n, E \cap M = \gamma n\}$$

Because the random bipartite graph G is constructed as the union of d independent perfect matchings, the probability that a selected configuration of vertices and edges does not violate the total matching constraints factors into the product of probabilities across each individual perfect matching.

Let λ be fugacity parameter. Then the expected number of total matchings is given by:

$$\begin{aligned} E[|\mathcal{T}_G^{\alpha,\beta,\gamma}|] = \lambda^{(\alpha+\beta+2\gamma)n} \frac{(n!)^2}{(\alpha n)!(\beta n)!(n-\alpha n-\gamma n)!(n-\beta n-\gamma n)!} \\ \cdot \sum_{\sum c_i = \gamma n} \prod_{i=1}^{d} \frac{(n-c_i-\beta n)!(n-c_i-\alpha n)!}{n!(n-c_i-\alpha n-\beta n)!c_i!} \end{aligned} \tag{3.1}$$

where the $\sum_{\sum c_i = \gamma n}$ evaluates over all possible valid combinations of the γn matching edges distributed across the d underlying perfect matchings, with c_i representing the number of matching edges contributed by the i-th perfect matching.

Lemma 3. A symmetric stationary point (local maximum) of the constrained first-moment objective ($E[|\mathcal{T}_G^{\alpha,\beta,\gamma}|]$) is given by $c_i = \frac{\gamma n}{d}$ $\forall i = 1, ..., d$.

Proof. The maximum value of $\mathbb{E}\left[|\mathcal{T}_G^{\alpha,\beta,\gamma}|\right]$ with respect to the variables c_i depends on maximising the combinatorial product within the summation:

$$\sum_{\sum c_i = \gamma n} \prod_{i=1}^{d} \frac{(n-c_i-\beta n)!(n-c_i-\alpha n)!}{n!(n-c_i-\alpha n-\beta n)!c_i!}$$

Applying Stirling's approximation to this term. Letting $f(x) = x \ln x$, we take the natural logarithm of the product to form an additive objective function. The logarithmic contribution of each individual c_i is captured by the function $g(c_i)$:

$$g(c_i) = f(n-c_i-\beta n) + f(n-c_i-\alpha n) - f(n-c_i-\alpha n-\beta n) - f(c_i)$$

Our goal is to maximise the sum of these contributions $\sum_{i=1}^{d} g(c_i)$ subject to the constraint: $\sum_{i=1}^{d} c_i = \gamma n$. We solve this constrained optimisation problem using the method

of Lagrange multipliers. Taking the first derivative of $g(c_i)$ with respect to c_i yields:

$$g'(c_i) = -[\ln(n-c_i-\beta n)+1]-[\ln(n-c_i-\alpha n)+1]+[\ln(n-c_i-\alpha n-\beta n)+1]-[\ln(c_i)+1]$$

$$g'(c_i) = \ln\left(\frac{n-c_i-\alpha n-\beta n}{c_i(n-c_i-\alpha n)(n-c_i-\beta n)}\right)-2$$

The Lagrange multiplier condition states that at a stationary point, the partial derivatives for all c_i must equate to the same constant multiplier μ:

$$g'(c_1) = g'(c_2) = \cdots = g'(c_d) = \mu$$

To determine whether the symmetric stationary point is a local maximum, we evaluate the second derivative $g''(c_i)$:

$$g''(c_i) = \left(\frac{1}{n-c_i-\beta n}-\frac{1}{n-c_i-\alpha n-\beta n}\right)+\left(\frac{1}{n-c_i-\alpha n}-\frac{1}{c_i}\right)$$

The first term is negative because $\alpha n \geq 0$. For graphs of degree $d \geq 3$, near the symmetric solution we have $c_i \leq \frac{\gamma n}{3}$. Given the total matching constraint $\alpha+\gamma \leq 1$, it follows that $c_i < \frac{n(1-\alpha)}{2}$ at the symmetric point $c_i^* = \frac{\gamma n}{d}$ because the feasibility condition states $\alpha+\gamma \leq 1$ we get

$$c_i^* \leq \frac{\gamma n}{3} \leq \frac{n(1-\alpha)}{3} < \frac{n(1-\alpha)}{2}$$

which guarantees the second term is also strictly negative. Since $g''(c_i^*) < 0$, the function g is strictly concave in a neighbourhood of the symmetric point. Consequently, the constrained objective $\sum_{i=1}^{d} g(c_i)$ has a local maximum at:

$$c_1 = c_2 = \cdots = c_d = \frac{\gamma n}{d}$$

Therefore, the symmetric allocation of matching edges is a local maximiser of the first-moment objective.

□

3.2.2 Setting Up the Stationary Point Equations

To obtain the exponential form of the rate function for the first moment, we evaluate the contribution at the symmetric local maximiser identified in Lemma 3. We use the symmetric contribution to define the exponential rate function for the subsequent stationary point analysis. Let T_{sym} denote the rate function evaluated at the symmetric local maximiser. This contribution scales exponentially as $T_{sym} = \exp(n \cdot \Phi_{sym})$. Because

our objective is to derive the continuous rate function Φ for the first moment, we bound the total summation S and evaluate the logarithmic limit as $n \to \infty$ in the neighbourhood of the symmetric root:

$$\frac{1}{n}\ln T_{sym} \leq \frac{1}{n}\ln S \leq \frac{1}{n}\ln(\text{poly(n)}) + \frac{1}{n}\ln T_{sym}$$

As $n \to \infty$ the polynomial error term $\frac{\ln(\text{poly(n)})}{n}$ goes to zero. Applying Stirling's approximation and substituting the symmetric stationary distribution $c_i = \frac{\gamma n}{d}$ into 3.1, in the neighbourhood of the symmetric point, the rate function is given by:

$$\mathbb{E}\left[|\mathcal{I}_G^{\alpha,\beta,\gamma}|\right] = \tilde{\Theta}\left(e^{n\cdot\Phi(\alpha,\beta,\gamma)}\right) \tag{3.2}$$

where $\Phi(\alpha,\beta,\gamma)$ is given by:

$$\begin{aligned}\Phi(\alpha,\beta,\gamma) = {} & (\alpha+\beta+2\gamma)\ln\lambda \\ & - f(\alpha) - f(\beta) - f(1-\alpha-\gamma) - f(1-\beta-\gamma) \\ & + d\left[f\left(1-\frac{\gamma}{d}-\alpha\right) + f\left(1-\frac{\gamma}{d}-\beta\right) - f\left(1-\frac{\gamma}{d}-\alpha-\beta\right) - f\left(\frac{\gamma}{d}\right)\right]\end{aligned} \tag{3.3}$$

where $f(x) = x\ln x$. Mossel, Weitz, and Wormald showed that structural shift occurs for computing the number of independent sets in a graph by analysing the stationary points of $\Phi(\alpha,\beta)$ [4]. Our function $\Phi(\alpha,\beta,\gamma)$ governs the total matchings in the probabilistic space satisfying $\alpha+\gamma \leq 1$, $\beta+\gamma \leq 1$ and $\alpha \geq 0, \beta \geq 0, \gamma \geq 0$, so we try to follow their analytical approach to identify the uniqueness region. To compute critical points of Φ, lets start by computing partial derivatives of the function Φ with respect to density parameters and then setting them to zero to get critical points:

$$\frac{\partial\Phi}{\partial\alpha} = \ln\lambda - \ln\alpha + \ln(1-\alpha-\gamma) + d\left[-\ln\left(1-\frac{\gamma}{d}-\alpha\right) + \ln\left(1-\frac{\gamma}{d}-\alpha-\beta\right)\right] \tag{3.4}$$

$$\frac{\partial\Phi}{\partial\beta} = \ln\lambda - \ln\beta + \ln(1-\beta-\gamma) + d\left[-\ln\left(1-\frac{\gamma}{d}-\beta\right) + \ln\left(1-\frac{\gamma}{d}-\alpha-\beta\right)\right] \tag{3.5}$$

$$\begin{aligned}\frac{\partial\Phi}{\partial\gamma} = {} & 2\ln\lambda + \ln(1-\alpha-\gamma) + \ln(1-\beta-\gamma) - \ln\left(1-\frac{\gamma}{d}-\alpha\right) \\ & - \ln\left(1-\frac{\gamma}{d}-\beta\right) + \ln\left(1-\frac{\gamma}{d}-\alpha-\beta\right) - \ln\left(\frac{\gamma}{d}\right) \\ & \frac{\partial\Phi}{\partial\alpha} = 0, \quad \frac{\partial\Phi}{\partial\beta} = 0, \quad \frac{\partial\Phi}{\partial\gamma} = 0\end{aligned} \tag{3.6}$$

We get the following equations for λ:

$$\lambda = \frac{x(1-\gamma/d-x)^d}{(1-x-\gamma)(1-\gamma/d-\alpha-\beta)^d} \text{ where x } \in \{\alpha,\beta\} \tag{3.7}$$

$$\lambda^2 = \frac{(1-\gamma/d-\alpha)(1-\gamma/d-\beta)(\gamma/d)}{(1-\alpha-\gamma)(1-\beta-\gamma)(1-\gamma/d-\alpha-\beta)} \tag{3.8}$$

The second derivatives:

$$\frac{\partial^2\Phi}{\partial\alpha^2} = -\frac{1}{\alpha} - \frac{1}{1-\alpha-\gamma} + \frac{d}{1-\frac{\gamma}{d}-\alpha} - \frac{d}{1-\frac{\gamma}{d}-\alpha-\beta} \tag{3.9}$$

$$\frac{\partial^2\Phi}{\partial\beta^2} = -\frac{1}{\beta} - \frac{1}{1-\beta-\gamma} + \frac{d}{1-\frac{\gamma}{d}-\beta} - \frac{d}{1-\frac{\gamma}{d}-\alpha-\beta} \tag{3.10}$$

$$\frac{\partial^2\Phi}{\partial\alpha\partial\beta} = -\frac{d}{1-\frac{\gamma}{d}-\alpha-\beta} \tag{3.11}$$

The region for which these equation holds is defined by $\alpha+\gamma \leq 1$ $\beta+\gamma \leq 1$ and $\alpha \geq 0, \beta \geq 0, \gamma \geq 0$.

From equation 3.4 and 3.5 for any γ in the feasible region evaluating at the unweighted $\lambda = 1$, the condition for stationary points satisfies the following:

$$\beta = f_\gamma(\alpha) \text{ and } \alpha = f_\gamma(\beta)$$

where the density function $f(x,\gamma)$ is given by:

$$f_\gamma(x) = \left(1-\frac{\gamma}{d}-x\right)\left(1-\left(\frac{x}{1-x-\gamma}\right)^{\frac{1}{d}}\right) \text{ where } x \in \{\alpha,\beta\}$$

Thus, at any stationary point, we require $\alpha = f_\gamma(f_\gamma(\alpha))$, bounded by the constraint $x+\gamma \leq 1, \alpha = \beta = x$.

To analyse the roots of the equation $\alpha = f(f(\alpha))$ specifically to find exactly at which degrees d, the symmetric root splits into multiple roots, we introduce change of variable.

$$y = \left(\frac{x}{1-x-\gamma}\right)^{1/d}$$

Rearranging to isolate x:

$$x = \frac{y^d(1-\gamma)}{1+y^d}$$

Expressing $f_\gamma(x)$ in terms of y, we get:

$$f(x) = \left(1-\frac{\gamma}{d}-x\right)(1-y)$$

for the first term we have, $1-x-\gamma = \frac{1-\gamma}{1+y^d}$

$$1-\frac{\gamma}{d}-x = \frac{1-\gamma}{1+y^d} + \gamma\left(\frac{d-1}{d}\right)$$

The f(x) is given by:

$$f(x) = \left[\frac{1-\gamma}{1+y^d} + \gamma\left(\frac{d-1}{d}\right)\right](1-y)$$

Now to evaluate f(f(x)), let $K = 1 - \frac{\gamma}{d} - x$.

$$K = \frac{1-\gamma}{1+y^d} + \gamma\left(\frac{d-1}{d}\right)$$

This implies that the density function $f(x)$ is: $f(x) = K(1-y) = K - Ky$

$$f(f(x)) = \left(1 - \frac{\gamma}{d} - f(x)\right)\left[1 - \left(\frac{f(x)}{1-f(x)-\gamma}\right)^{1/d}\right]$$

$$f(f(x)) = (x+Ky)\left[1 - \left(\frac{f(x)}{1-\gamma-f(x)}\right)^{1/d}\right] \tag{3.12}$$

To analyse the roots of $f(f(x)) = x$ i.e. $x = (x+Ky)\left[1 - \left(\frac{f(x)}{1-\gamma-f(x)}\right)^{1/d}\right]$ we get the following simplified form:

$$\left(\frac{Ky}{x+Ky}\right)^d = \frac{f(x)}{1-\gamma-f(x)} \tag{3.13}$$

To determine the specific values of x (and consequently y) for which the stationary point equation holds, we take the natural logarithm of both sides to construct a continuous function $g(y)$:

$$g(y) = d\ln(Ky) - d\ln(x+Ky) - \ln(f(x)) + \ln(1-\gamma-f(x))$$

3.2.3 Stationary Points and Local Stability of the Total Matching Model

The moment analysis above provides a system of equations describing the stationary points of the first-moment rate function $\Phi(\alpha,\beta,\gamma)$. These equations can be used to identify symmetric and asymmetric phases of the total matching model on random d-regular bipartite graphs. However, the existence of multiple stationary points or the local instability of a symmetric stationary point does not imply a proof of approximate counting hardness. Therefore, we state the conclusion of the present analysis in terms of the stationary point structure and local stability.

Let $G(V_1 \cup V_2, E)$ be a random d-regular bipartite graph with $|V_1| = |V_2| = n$, and a matching density γ for which the feasible region

$$\alpha \geq 0, \qquad \beta \geq 0, \qquad \gamma \geq 0, \qquad \alpha+\gamma \leq 1, \qquad \beta+\gamma \leq 1$$

is non-empty. For $\lambda = 1$, the stationary-point equations derived from (3.4) and (3.5) can be written as

$$\beta = f_\gamma(\alpha), \qquad \alpha = f_\gamma(\beta),$$

where

$$f_\gamma(x) = \left(1 - \frac{\gamma}{d} - x\right)\left(1 - \left(\frac{x}{1-x-\gamma}\right)^{1/d}\right) \tag{3.14}$$

Consequently, any symmetric stationary point satisfies

$$\alpha = \beta = x^*,$$

and hence

$$x^* = f_\gamma(x^*) \tag{3.15}$$

We have

$$y(x) = \left(\frac{x}{1-x-\gamma}\right)^{1/d}$$

Differentiating gives:

$$y'(x) = \frac{y(x)(1-\gamma)}{d\,x(1-x-\gamma)} \tag{3.16}$$

Differentiating f_γ gives:

$$f'_\gamma(x) = -(1-y(x)) - \left(1 - \frac{\gamma}{d} - x\right)\frac{y(x)(1-\gamma)}{d\,x(1-x-\gamma)} \tag{3.17}$$

At a symmetric fixed point x^*, equation (3.15) implies:

$$x^* = \left(1 - \frac{\gamma}{d} - x^*\right)(1-y^*),$$

where

$$y^* = \left(\frac{x^*}{1-x^*-\gamma}\right)^{1/d}$$

substituting into (3.17) gives

$$f'_\gamma(x^*) = -(1-y^*) - \frac{y^*(1-\gamma)}{d(1-y^*)(1-x^*-\gamma)} \tag{3.18}$$

This expression determines the local stability of the fixed point. Consider the 2D map

$$F(\alpha,\beta) = \big(f_\gamma(\beta), f_\gamma(\alpha)\big).$$

At the symmetric stationary point (x^*,x^*), the Jacobian is given by:

$$DF(x^*,x^*) = \begin{pmatrix} 0 & f'_\gamma(x^*) \\ f'_\gamma(x^*) & 0 \end{pmatrix} \tag{3.19}$$

The symmetric fixed point is locally stable whenever

$$|f'_\gamma(x^*)| < 1 \tag{3.20}$$

and loses its stability when

$$|f'_\gamma(x^*)| > 1 \tag{3.21}$$

Since $f'_\gamma(x^*) < 0$, the instability condition can be written as

$$f'_\gamma(x^*) < -1 \tag{3.22}$$

Substituting 3.18, this is equivalent to

$$\frac{1-\gamma}{d(1-y^*)(1-x^*-\gamma)} > 1 \tag{3.23}$$

Equation 3.23 gives the linear stability threshold for the symmetric fixed point.

Consequence for phase-transition analysis

The 3.23 inequality provides a criterion for identifying parameter values at which the symmetric solution may cease to be locally stable. For fixed (d,γ), we solve 3.15 for x^*, evaluate 3.18, and check whether 3.20 or 3.22 holds. This gives a procedure for computing transition points. A transition value $d_c(\gamma)$, when it exists, can be defined by

$$|f'_\gamma(x^*)| = 1 \tag{3.24}$$

together with

$$x^* = f_\gamma(x^*) \tag{3.25}$$

The numerical analysis in Section 3.3 explores this stationary point structure. The condition $|f'_\gamma(x^*)| < 1$ implies stability of the fixed-point equations derived from the first-moment function.

To establish hardness following the approach of Sly [5]: construct a gadget with distinguishable phases and represent copies so that phase assignments represent a solution to an NP-hard problem (MAX-CUT). For total matchings, the matching constraint introduces dependencies absent in independent-set models. In particular, if two gadget copies are connected by cross-edges, we cannot assign independent multiplicative weights to each cross-edge. A matching is a set of edges with no shared endpoints, so the contribution of one cross-edge depends on the occupancy of others incident to the same vertices. Thus we must evaluate the partition function of the cross-edge structure while enforcing:

1. no endpoint of a matching edge is reused, and

2. no two matching edges share a vertex.

We tried the simplified expression

$$M_{u,v} = 1 - q_u q_v + (1 - q_u)(1 - q_v) = 2 - q_u - q_v$$

however it does not correctly account for these dependencies when multiple cross edges share vertices.

For the random d-regular bipartite total-matching model, every interior symmetric stationary point x^* satisfies

$$x^* = f_\gamma(x^*),$$

where f_γ is defined in 3.14. The Jacobian of the map has eigenvalues $\pm f'_\gamma(x^*)$. So the symmetric stationary point is locally stable when $|f'_\gamma(x^*)| < 1$ and unstable when $|f'_\gamma(x^*)| > 1$. The critical parameter values are determined by 3.24 and 3.25.

3.3 Numerical Analysis

In this section we investigate the roots of the equation g(y)=0. A single root corresponds to a unique solution of the stationary point equation within the numerical domain considered, whereas the presence of multiple roots indicates the existence of multiple stationary point solutions. These stationary points indicate the possibility of a phase transition and motivate further investigation of thermodynamic non-uniqueness. The observation of three roots does not by itself prove that the Gibbs measure is non-unique or that approximate counting is computationally hard. The numerical results below should be interpreted as evidence for a change in the stationary-point structure.

Because the total matchings model is a hybrid structure governed by the independent set densities (α, β) and the matching edge density γ, the degree d_c at which we observe three roots is a function of γ. We evaluate $g(y) = 0$ numerically for three distinct structural regimes: an independent set-dominated regime ($\gamma = 0.1$), ($\gamma = 0.5$), and a matching-dominated regime ($\gamma = 0.9$).

For $\gamma = 0.1$, the numerical results show a transition from one detected root to three detected roots between d=5 and d=6. This indicates a change in the structure of the stationary point equations in the independent set-dominated regime and is consistent with the phase-transition behaviour observed in the hardcore model.

Table 3.1: Analysis of Function $g(y) = 0$ for $\gamma = 0.1$

Degree (d)	**Gamma** (γ)	**Number of Roots**	**Roots** (y)
3	0.1	1	0.69479
4	0.1	1	0.73563
5	0.1	1	0.76477
6	0.1	3	0.67030, 0.78693, 0.89055

Table 3.2: Analysis of Function $g(y) = 0$ for $\gamma = 0.5$

Degree (d)	**Gamma** (γ)	**Number of Roots**	**Roots** (y)
6	0.5	1	0.8374
7	0.5	1	0.84982
8	0.5	1	0.85999
9	0.5	1	0.86852
10	0.5	1	0.87583
11	0.5	3	0.81831, 0.88219, 0.9372

Table 3.3: Analysis of Function $g(y) = 0$ for $\gamma = 0.9$

Degree (d)	**Gamma** (γ)	**Number of Roots**	**Roots** (y)
3	0.9	1	0.93176
4	0.9	1	0.94003
6	0.9	1	0.94795
7	0.9	1	0.95039
8	0.9	1	0.95236
9	0.9	1	0.95402
10	0.9	1	0.95546
11	0.9	1	0.95674
12	0.9	1	0.95789
13	0.9	1	0.95895
14	0.9	1	0.95992
15	0.9	1	0.96082
60	0.9	3	0.96663, 0.97819, 0.98788

Table 3.2 illustrates that introducing a higher density of matchings changes the degree thresholds for which transition occurs. A graph of degree $d = 6$ is no longer sufficient to observe multiple roots. The first degree at which three stationary-point roots are detected is $d_c = 11$.

Now we evaluate the system at the upper bound of the feasible region, where the total matching configuration is almost entirely dominated by matching edges, leaving independent sets as a marginal constraint ($\gamma = 0.9$). In this regime, three stationary point roots are first detected at $d_c = 60$.

These results indicate that the stationary-point structure of the total-matching model depends strongly on the matching density γ, with multiple stationary-point solutions appearing at larger degrees as γ increases.

Chapter 4

Final Remarks and Future Horizons

4.1 Conclusions

In this dissertation, we studied the computational complexity and structural properties of the #TOTAL MATCHINGS problem.

Our study began by proving the exact counting hardness. By introducing a parameterised intermediate problem, #TOTAL MATCHINGS(i, j), we were able to isolate specific structural densities. Through a Turing (multicall) reduction from the known #P-hard problem (#INDSET), utilising polynomial interpolation technique, we proved that evaluating the #TOTAL MATCHINGS partition function is #P-hard for general graphs. We also proved very important lemmas for showing the invertibility of the coefficient matrix.

In the approximate regime, we adapted the phase transition frameworks introduced by Sly, Mossel, Weitz, and Wormald [4, 5, 7] to investigate the behaviour of total matchings on random bounded degree bipartite graphs. Our initial attempts to establish approximation hardness utilising simpler graph modifications such as linear chains and tree structures proved insufficient, as they failed to put more weight on large independent sets or failed to reduce the error term to zero, as the parameters are increased $(n \to \infty)$. For the moment analysis we introduced an additional density parameter (γ) describing the proportion of matching edges, leading to a first-moment rate function $\Phi(\alpha, \beta, \gamma)$. From this rate function we derived the stationary-point equations and a local stability criterion for the symmetric solution.

The numerical analysis results at the end of chapter 3, indicate that the stationary-point structure depends strongly on the matching density γ. In particular, increasing γ shifts the parameter regime in which multiple stationary-point solutions emerge. These

observations suggest that the interaction between independent-set and matching constraints may produce a phase diagram that differs from that of the hard-core model alone. This work provides an analytical framework for investigating the hardness of approximate counting, rather than a complete complexity classification in the approximate regime.

This work demonstrates that total matchings represent a unique, highly dynamic graph structure, where the boundaries of algorithmic tractability are dictated by the internal balance between its component sub-structures.

4.2 Future Directions

The results achieved in this project can be further improved. This opens several promising ways for future research in computational complexity and statistical physics.

Novel Structural Gadgets and Boundary Analysis: Our current numerical analysis for evaluating the phase transition thresholds is based on previous methodologies designed for independent sets. Future work involves identifying a completely novel structural gadget specifically tailored to the unique constraints of total matchings to more accurately map and prove the exact hardness boundaries. This would formally prove the existence of an FPTAS or FPRAS below the established critical thresholds.

Explicit Parameterisation of Matching Constraints: Furthermore, in analysing the approximate hardness for bounded-degree graphs, our adaptation of Sly's approach did not utilise an explicit additional parameter for the matchings. Instead, we relied on a simplified structural equivalence, treating two adjacent free vertices as a sufficient condition for a valid matching edge. This framework can be significantly improved by introducing a parameter that explicitly represents the marginal probability that a given vertex is covered by a matching. Expanding the model to incorporate this matching probability would allow for a much more rigorous analysis of the thermodynamic phase transitions. Determining whether an appropriate bounded degree interaction can favour opposite phases and thereby encode the MAX-CUT [5], is an important step.

Glauber Dynamics and Mixing Times: This involves analysing the mixing time of local Markov chains (such as Glauber dynamics) over the total matchings configuration space. Establishing slow mixing in a non-unique regime would provide a way to understand the computational behaviour of the model and could help connect the stationary point analysis to algorithmic hardness.

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Appendix A

First appendix

A.1 Numerical Analysis for g(y)

```
import numpy as np
from scipy.optimize import root_scalar

def analyze_gy_roots(d, gamma):
    """
    Attempts to locate valid roots of the logarithmic function
    g(y)=0 in the interval (0,1) using a uniform grid
    and sign-change detection.
    """
    def g(y):
        # restriction for y
        if y <= 0 or y >= 1:
            return np.nan

        x = (y**d * (1 - gamma)) / (1 + y**d)
        K = (1 - gamma) / (1 + y**d) + gamma * ((d - 1) / d)
        f_x = K * (1 - y)

        if K * y <= 0 or x + K * y <= 0 or f_x <= 0
        or 1 - gamma - f_x <= 0:
            return np.nan
```

```
        # Evaluate
        term1 = d * np.log(K * y)
        term2 = d * np.log(x + K * y)
        term3 = np.log(f_x)
        term4 = np.log(1 - gamma - f_x)

        return term1 - term2 - term3 + term4

    y_vals = np.linspace(1e-5, 1 - 1e-5, 5000)
    g_vals = [g(y) for y in y_vals]

    roots = []
    for i in range(len(y_vals) - 1):
        g1, g2 = g_vals[i], g_vals[i+1]

        # if a sign change occurs --
        # root exists between these two points
        if not np.isnan(g1) and
        not np.isnan(g2) and g1 * g2 <= 0:
            res = root_scalar(g, bracket=[y_vals[i],
                y_vals[i+1]], method='brentq')
            if res.converged:
                root_val = round(res.root, 5)
                if root_val not in roots:
                    roots.append(root_val)

    return roots

# testing for various values of gamma
test_cases = [
    {"d": 3, "gamma": 0.1},
    {"d": 4, "gamma": 0.1},
    {"d": 5, "gamma": 0.1},
    {"d": 6, "gamma": 0.1},
    {"d": 7, "gamma": 0.1},
```

```
    {"d": 8, "gamma": 0.1},
    {"d": 9, "gamma": 0.1},
    {"d": 10, "gamma": 0.1},
    {"d": 11, "gamma": 0.1},
    {"d": 12, "gamma": 0.1},
    {"d": 13, "gamma": 0.1},
    {"d": 14, "gamma": 0.1},
    {"d": 15, "gamma": 0.1},
    {"d": 60, "gamma": 0.1},
]

print("Root Analysis using Logarithmic Function g(y) = 0")
print("-" * 55)
for case in test_cases:
    d = case["d"]
    gamma = case["gamma"]
    roots = analyze_gy_roots(d, gamma)

    print(f"Degree d = {d}, Gamma = {gamma}")
    print(f"Found {len(roots)} root(s) for y: {roots}")
    print("-" * 55)
```